



# SC1006 Computer Organization and Architecture

## Briefing (Second Half)

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# Contact Info

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# Expectation alignment

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- **Online Lecture Video**

- Cover concepts.
- There are some overloading in the first week (Week 8) to get students up to speed with content required for the Lab sessions and Lab quiz.
- Online quiz questions will be provided, students may need to do some self investigation when deriving the answers. There will similar MCQs in the final exam.
- Helping notes will be supplied to aid in understanding how the answers are derived, whenever necessary.
- Detail notes on the lecture slides are provided in the course site lecture folder.
- Updated Common Terms/Definitions/Explanations are compiled.

- **Physical Review Sessions**

- Review of important/complex concepts (not all sub-topics will be covered) and selected online quiz questions
- Supplementary topics, review of PYP questions, optional tutorial questions, frequently-asked questions from students, etc

# Expectation alignment

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- **Tutorials and Labs**

- Applications of concepts. As such, there will be gaps between concepts learnt and tutorial solution so do not expect to be able to lift answers directly from the lecture notes.
- Do not expect a lot of hand holding or step by step instructions.  
**Queries are welcomed and encouraged.**

- **Labs Specific**

- Watch the video “Lab3 Briefing” before coming to lab3.
- Watch the video “Lab4 Briefing” before coming to lab4.

# Pre-recorded lecture video

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- Available online in NTULearn.
- A separate power point show file is available to quiz on the weekly content to allow students to self-assess their understanding of the subject material.
- Textual elaboration of how answer is derived is included in the pps file, when needed.
- View these before attending on-site review sessions.

# Review & Q/A sessions

| Week             | Mon              | Fri                |
|------------------|------------------|--------------------|
| 8 (07/10-11/10)  | (mid-term exam)  | Q/A (on-line)      |
| 9 (14/10-18/10)  | Review (on-site) | Review (on-site)   |
| 10 (21/10-25/10) | Review (on-site) | Review (on-site)   |
| 11 (28/10-01/11) | Q/A (on-line)    | Q/A (on-site/line) |
| 12 (4/11-08/11)  | Review (on-site) | Review (on-site)   |
| 13 (11/11-15/11) | Review (on-site) | Q/A (on-site/line) |

## Links for Q/A:

### Fri (Wks 8,11,13):

<https://ntu-sg.zoom.us/j/87057751587?pwd=SHaTDBluFK2gHbvUNasUGOHZdtpCcU.1>

Meeting ID: 870 5775 1587

Passcode: 737992

### Mon (Wk 11)

<https://ntu-sg.zoom.us/j/89047289435?pwd=WXJiiBmG98QXKh2K3FX87U4ix6yz0e.1>

Meeting ID: 890 4728 9435

Passcode: 799556

# Tutorial

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Tutorials for Part 2 starts from Week 9 (next week)

Tutorial #5, #6 and #7 will be structured around a piece of document named “Case Study Note”.

- The case study note is a collection of hardware technical information; the purpose is to emulate information that is available in product datasheets and user manual.
- Part of the tutorial objective is to exercise students on the process of identifying and retrieving relevant information from product manuals to tackle a problem.

# Lab

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- **Lab3**

- Arduino Development Platform.
- Relevant materials
  - Week8 lectures on Computer Organisation Overview and Signal Chain Sub-System, Tutorial #5.
- Watch the Video on “SC1006-Lab3 Briefing” in the Lab3 folder in NTULearn, before coming to the Lab.
- Involves hardware and dealing with C programs.
  - “C Programming” aspect of the lab involve mainly compilation and modifying some variables/parameters in the sample C code provided.

- **Lab4**

- Cache and Virtual Memory Systems
- Relevant materials
  - Week10 lectures on Cache and Virtual Memory System
  - Tutorial #8
    - Tutorial 8 video uploaded in NTULearn to aid early preparation.



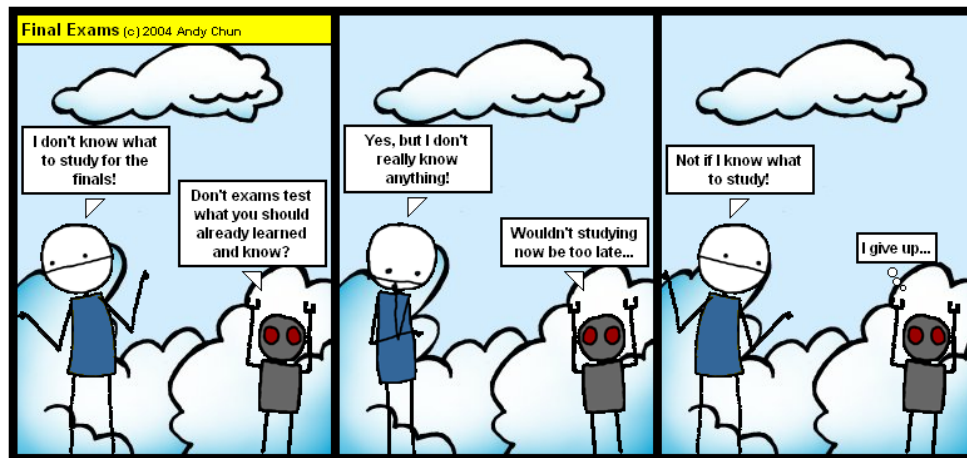
# Lab Quiz

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- Open book. 10 mins. 7 MCQ. Online.
- Content that will be quizzed includes
  - Direct knowledge related to task performed in the lab.
  - Lab task related materials covered in lecture and tutorial; e.g. some of the Lab3 tasks deal with Asynchronous and Synchronous Serial interface topics, which is discussed in week8 lectures under Signal Chain Sub system. Therefore, expect questions related to these topics.
- Lab3 and Lab4 manuals are uploaded to NTULearn, students may download them to browse through the tasks involved.

# Final exam (for Part 2 only)

- Duration: 1 hour
- Closed Book
- In-Person Exam
- Mixture of MCQs and Very-Short-Answer Questions
- Second Half Content (After Recess Week): Includes Lecture, tutorial & lab content.
- “Comms and UI” related questions will only appear in MCQs.
- Will brief further during one of the review sessions.



# Additional references

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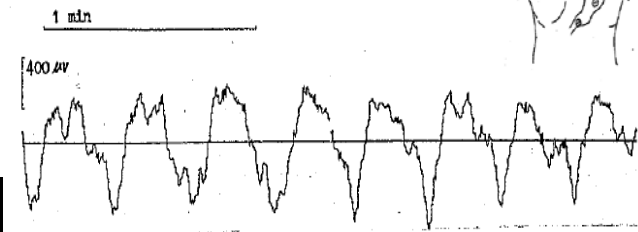
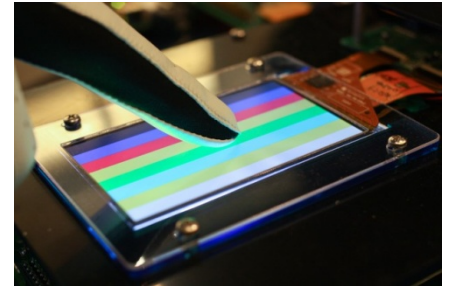
- Linda Null and Julia Lobur, “The Essentials of Computer Organization and Architecture”, Jones & Bartlett Learning, 3<sup>rd</sup> Edition, 2012
- Randal E. Bryant and David O’Hallaron, “Computer Systems – A Programmer’s Perspective”, Prentice Hall, 2<sup>nd</sup> Edition, 2011.

Part 2 covers diversified topics and is from many other sources. Therefore, you may need to search for additional on-line resources whenever necessary.

# Major Related Systems/modules Developed for Companies

## Use of processors, memory & interfaces

- Person ReID for airports
- Quality Inspection of PDAs/HPs
- Perceptual Quality Metric for Video
- Edge Coder for Video Compression
- MP3 Player
- Real-time On-line Video Object Segmentation
- MPEG-4 Videophone with Object Segmentation & Error Resilience
- H.324 Video Conferencing System
- Glaucoma detection as a clinical aid
- Electrogastrographic (EGG) Detection & Analysis as Diagnosis Aid





*Questions, thinking questions, critically  
thinking questions...*

Teaching is not about  
providing answers.

It is about encouraging questions.

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SINGAPORE



# Extended Sequential Logic for Synchronous Circuit Optimization and Its Applications

Pramod Kumar Meher, *Senior Member, IEEE*

**Abstract**—In this paper, we present a new approach for the extension of sequential logic functionality of  $D$  flip-flop in order to perform an additional Boolean function simultaneously along with its usual bit-storage function. We show that a combinational function of the form  $(a \cdot b)$ ,  $(a + b)$ ,  $(a + \bar{b})$ , or  $(\bar{a} \cdot b)$  which occurs frequently in a feedforward path with a  $D$  flip-flop could be implemented efficiently by a  $D$  flip-flop with RESET or SET provision. Similarly,  $(a \oplus b)$  or  $((a \cdot b) \oplus c)$  in the feedback loop with a  $D$  flip-flop could be implemented by a  $T$  flip-flop by suitable modification of the clock. The use of such extended sequential logic is found to result in a significant reduction in critical path and saving in area complexity over the direct implementation. Moreover, we present a simple approach for the construction of CMOS  $T$  flip-flop by modification of clock signal of  $D$  flip-flop, which is found to be more efficient than the  $T$  flip-flop derived from  $JK$  flip-flop. The extended sequential logic is used for the implementation of finite-field multiplication over  $GF(2^m)$  and carry-save addition of real numbers. In both these cases, the use of extended logic is found to offer a substantial saving in area and time complexity over the conventional implementations.

vice versa. The basic Boolean function in a feedforward path could be  $(a \cdot b)$ ,  $(a + b)$ ,  $(a \oplus b)$ ,  $(a + \bar{b})$ , or  $(\bar{a} \cdot b)$ . Except  $(a \oplus b)$ , none of these functions appear to be practically relevant to be used in the feedback loop with a  $D$  flip-flop, since the output gets stuck at fixed output state on particular input conditions. Some typical forms of extended sequential logic of practical relevance which we have investigated and which we aim at presenting in this paper are as follows:

- 1)  $(a \cdot b)$  followed by  $D$  flip-flop;
- 2)  $(a + b)$  followed by  $D$  flip-flop;
- 3)  $(a + \bar{b})$  or  $(\bar{a} \cdot b)$  followed by  $D$  flip-flop;
- 4)  $(a \oplus b)$  followed by  $D$  flip-flop;
- 5)  $(a \oplus b)$  in feedback loop with  $D$  flip-flop;
- 6)  $((a \cdot b) \oplus c)$  in feedback loop with  $D$  flip-flop, where  $a$ ,  $b$ , and  $c$  are Boolean variables.

In a recent paper, we have shown that an XOR gate in the feedback loop with a  $D$  flip-flop could be replaced by a

# “WHW” Questions in Thinking & Re-thinking

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WHAT?

The problem we try to solve

HOW?

The methodology used

WHY (or WHY NOT)?

Do not believe until being convinced

Think critically

Ask questions

